MIS 201 · WEEK 7 · MONDAY 11/2

Ch. 7: Transportation:
The Empty Driver's Seat

Part 1. How a car learns to see and to decide, and where it still breaks.

Part 2. Waymo: driverless taxis in 14 U.S. cities.

Wednesday: you pitch Bellingham's city council for a robotaxi company.

This week: due Friday 11/6

DUE FRIDAY

WHAT

Lab 7 writeup and the Chapter 7 quiz.

WHERE

Both on Canvas.

WHEN

Friday 11/6, 11:59 p.m.

BEFORE WEDNESDAY

Skim the debate records on the Lab 7 page.

Hang on to your answer to today's question 5. You'll use it Wednesday.

Today

Two rides, and the loop
How it sees: neural networks, CNNs, sensors
How it decides: the score sheet
Practice worlds and the long tail
Where the human stands
Try it

PART 2 · THE WAYMO CASE · ~50 MIN

The story, and the World Model clips
How the Waymo Driver works
Who wins, who loses, what breaks
Discuss in groups, then as a class
Keep, fix, or drop

BY THE END, YOU CAN

- Name the five steps a self-driving car repeats
- Say how a neural network learns, in one sentence
- Explain why the score sheet is a business decision
- Explain why rare cases are the hard ones
- Say where the human is in any AI system

PART 1 · CHAPTER 7 · ABOUT 50 MINUTES

AI with a body: where is the human?

When AI gets a body, a mistake isn’t a bad sentence. It’s a two-ton car.

The hard part isn’t the driving. It’s the seeing and the rare surprises. And trust comes down to one question: where is the human?

Two rides, one difference



Ride one: AI suggests a route. A person decides.



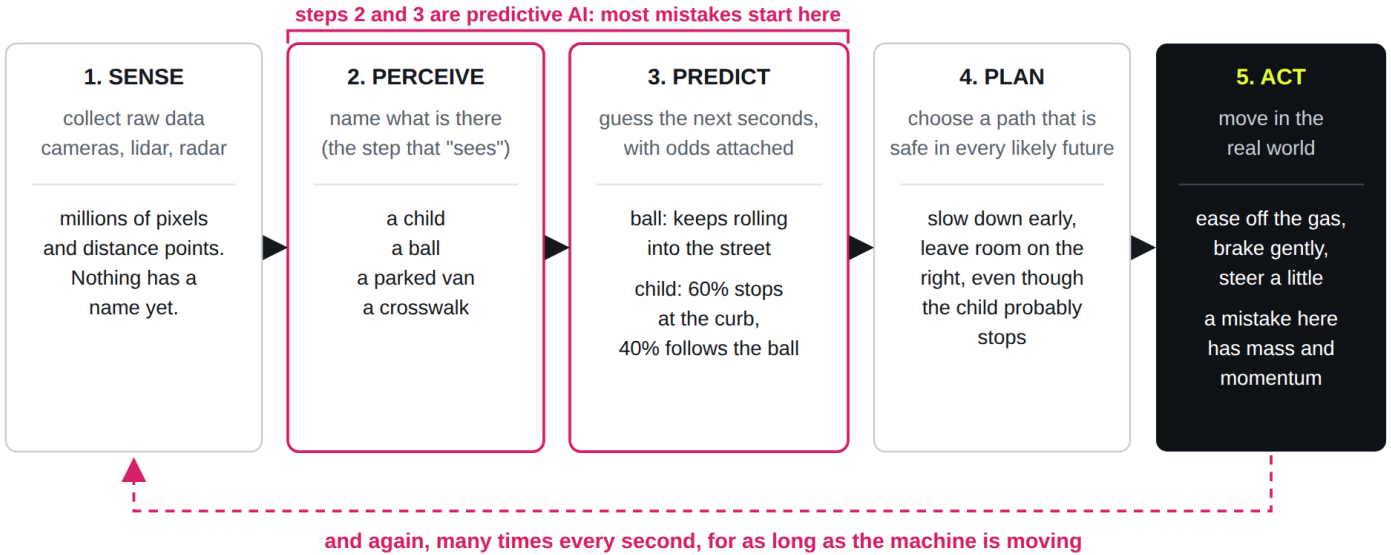
Ride two: nobody in the driver's seat. On purpose.

Same transportation industry. AI in both. The difference is where the human stands.

Photos: Jason Crotty; Daniel L. Lu. CC BY 4.0, via Wikimedia Commons.

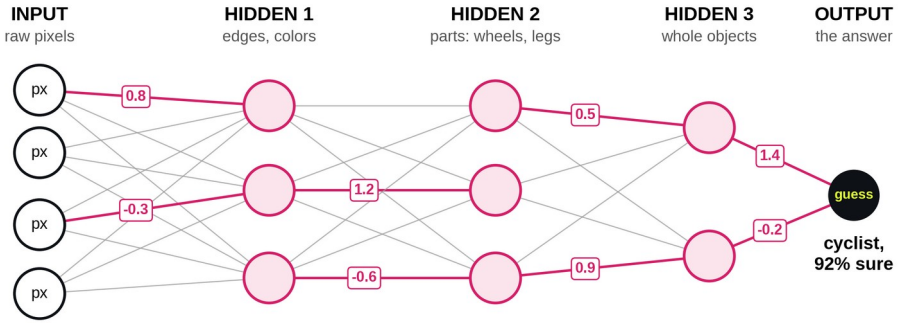
The loop a self-driving car repeats

Embodied AI = AI with a body. It senses the world and moves in it. A robot vacuum counts.



Steering is easy. Steps 2 and 3, naming things and guessing what they'll do, are hard.

How a car learns to see: a neural network



Each line is a weight: a number for how much one unit counts toward the next.
Training adjusts these numbers. Nobody programs what each layer looks for.
"Deep" means many hidden layers. Weights shown are made up; real networks have billions.

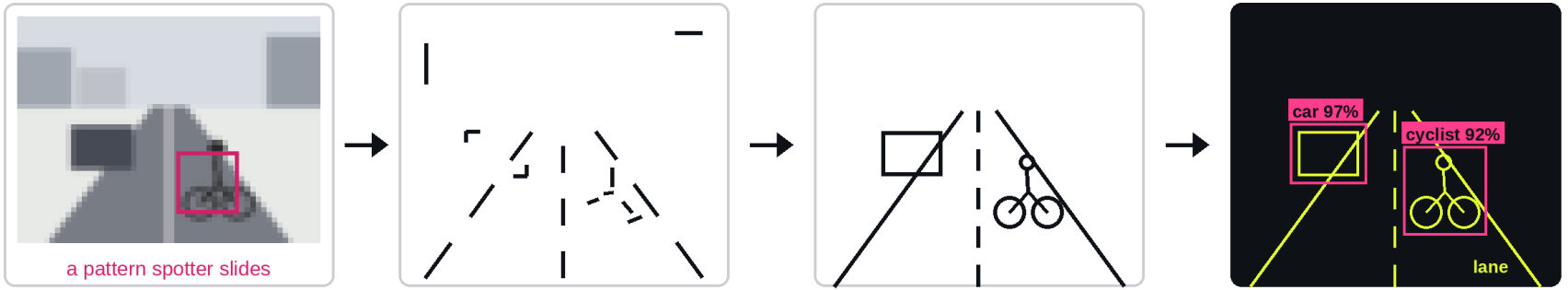
THINK OF IT AS A STACK OF DIALS

1. Show it a photo labeled “cyclist.”
2. It guesses.
3. Nudge every dial so the guess is a little less wrong. Repeat millions of times.

Neural network = layers of tiny math units, connected by numbers that get adjusted during training. It learns from examples, not from rules someone wrote. **Neural network** is the machine. **Deep learning** is a neural network with a lot of layers.

Two catches. It can’t tell you why it decided. And it only knows what it has seen.

Example: How a Neural Work Reads a photo: small spotters that slide



camera view
just numbers

layer 1: edges
lane lines, curbs, corners

layer 2: shapes
wheels, a box, a road edge

layer 3: objects
cyclist, car, lane

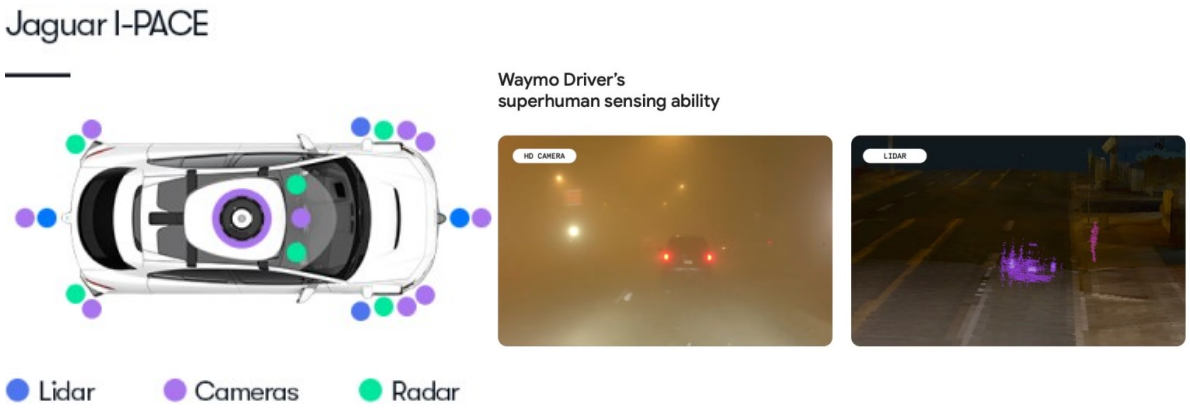
Many times a second, from every camera. Each layer's spotters are learned from examples, not written by a person.

CNN (convolutional neural network) = a neural network built for images. It slides small pattern spotters across a photo: edges first, then shapes, then whole objects. Nobody tells it what to look for.

'Convolutional' means sliding.

Your phone's face unlock runs one.

How does a car see? Three sensors



CAMERA

Sees color, signs, lights. Bad in the dark.

LIDAR (A LASER)

Exact distance, day or night. Bad in fog.

RADAR

Speed, in any weather. Blurry.

What lidar sees: no color, no words. Photo: Daniel L. Lu, CC BY 4.0.

When they disagree, who wins? People pick the tie-breaker.

How a Neural Network learns to decide: reinforcement learning

Reinforcement learning = learning by trial and error, with points. It tries something, gets points, and does more of whatever scored well. Like training a dog with treats.

A MADE-UP SCORE SHEET FOR SELF-DRIVING CAR (positive = good; negative = bad)

Complete the turn	**+10**
Close call (almost hit a cyclist)	**–50**
Each second spent waiting	**–1**
Crash	**–1000**



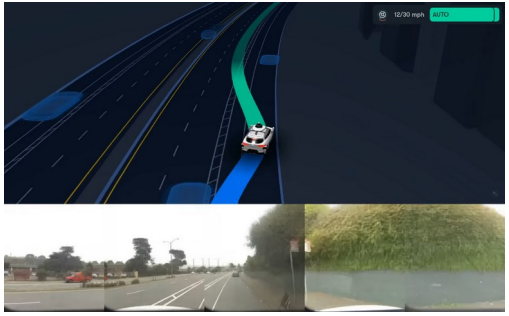
Photo: Veterinarian Pet Alliance, CC BY 4.0.

It learns what you scored, not what you meant. Someone picks these numbers.

Simulation and world models: practice in worlds that never happened



A flight simulator. Photo: Ilari Kousa, CC BY-SA 4.0, via Wikimedia Commons.



A Waymo World Model. Photo: https://waymo.com/blog/2026/02/the-waymo-world-model-a-new-frontier-for-autonomous-driving-simulation/

SIMULATION

A fake world where crashing costs nothing. Pilots train this way.

WORLD MODEL

AI that can invent new scenes and scenarios from what it learned watching video.

THE CATCH

It can only remix what it has seen. Practice isn’t proof.

The long tail: rare stuff is where it breaks



Photo: Retired electrician, CC0, via Wikimedia Commons.

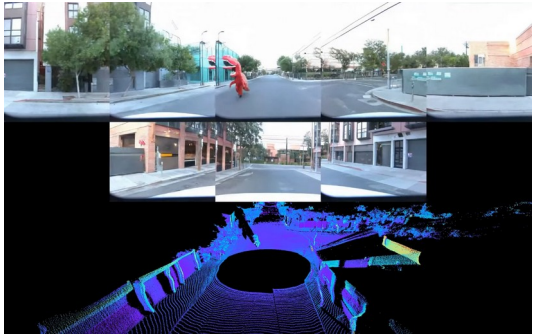
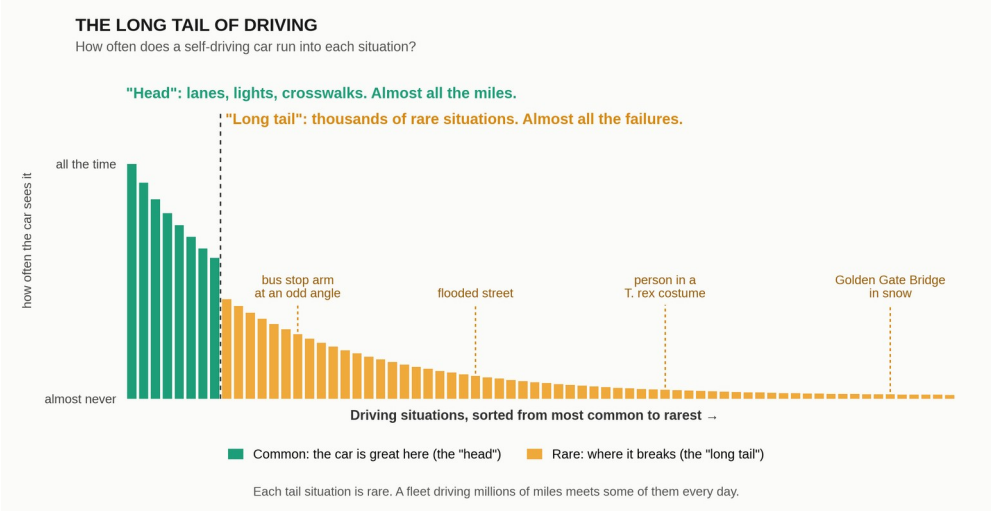


Photo: Waymo World Model of Dinosaur Crossing Street

Pedestrian, cyclist, or dinosaur?

You know instantly. A neural network that never saw one has to guess, and it guesses confidently.

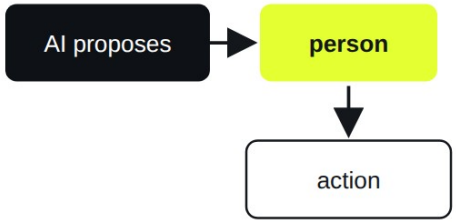


Right 99.99% of the time still means something goes wrong every day, at scale.

https://waymo.com/blog/2026/02/the-waymo-world-model-a-new-frontier-for-autonomous-driving-simulation/

Where the human stands: in, on, or out of the loop

IN THE LOOP

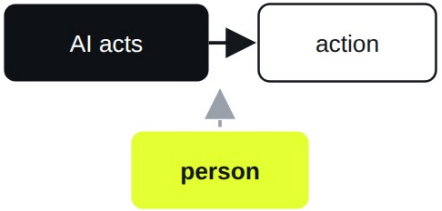


nothing happens without sign-off

Flyways. Every Cure. The human is the gate.

slow, and it holds

ON THE LOOP



watches, ready to grab control

a driver "supervising" a car that mostly drives itself

looks safest. often the worst seat:
attention decays, the handoff comes mid-crisis

OUT OF THE LOOP



no person positioned to
intervene in the moment

Waymo, on purpose

the bet: removing the unreliable human
beats keeping a badly placed one

The middle seat is a trap. People stop paying attention, then get handed the wheel at the worst moment.

Responsible AI: Four Questions to Answer

Responsible AI = building and running AI so that someone can check it, limit it, and answer for it when it goes wrong.

Can AI explain a mistake?

No, so the company needs testing, records, and replays.

Does AI know where it's safe?

Its approved zone: roads, speeds, weather, hours. Outside it, stop.

Who keeps the score?

Whoever counts the crashes holds the power.

Who answers when AI is wrong?

There's no driver to ticket. The company, the city, or both.

Wednesday, the city council will ask you all four.

Try it: name the idea

PRACTICE FOR FRIDAY

For each one: neural network, CNN, reinforcement learning, world model, or long tail?

1. A car's software gets points for finishing a turn and loses points for close calls. After millions of tries, it merges more smoothly.
2. An engineer types "a mattress falls off a truck at night in the rain," and the system creates fake camera video of it for the car to practice on.
3. A car handles a million ordinary intersections without trouble, then freezes when a parade float rolls through.
4. Software slides a small window across each camera frame, spotting edges first, then wheels, then "cyclist."
5. Shown millions of labeled photos, a system adjusts its numbers after each guess until it gets "stop sign" right. Nobody can point to which numbers mean "stop sign."

Try it: the answers

PRACTICE FOR FRIDAY

1. **Reinforcement learning.** Points, not labeled answers. It learns what you scored, not what you meant.
2. **World model.** It invents a scene that never happened. Practice, not proof..
3. **Long tail.** Great on the common stuff, lost on the rare stuff.
4. **CNN.** The sliding spotters are the giveaway.
5. **Neural network.** A CNN is one too; what makes item 4 a CNN is the sliding.

Seeing learns from labeled answers. Deciding learns from points. Practice happens in fake worlds. Failure lives in the tail.

PART 2 · THE CASE · ABOUT 50 MINUTES

Need a break?? Go for it,

Waymo: nobody in the driver's seat

Driverless taxis were always "five years away."
Then, one city at a time, they weren't.

Everything from Part 1 is inside this one car.

Watch: scenes that never happened

VIDEO



WATCH FOR

1. Which rare situation is each clip practicing?
2. None of this is real footage. Can you tell?
3. Who checked that it’s realistic?

Waymo World Model · Waymo · February 2026

Waymo, by the numbers

CASE



Photo: Daniel L. Lu, CC BY 4.0, via Wikimedia Commons.

500K+ paid rides a week (company's count)

14 U.S. cities, as of September 2026

271M miles with no driver (company's count)

Who wins, and who loses?

CASE

The doctor (NPR, Dec 2025)

90%+ fewer of worst crashes

A trauma surgeon read Waymo's data and saw over 90% fewer of the worst crashes he treats. He calls it a public health breakthrough.

WINS: RIDERS

Safer w/ no driver

Some women and disabled riders say they feel safer with no driver.

LOSES: DRIVERS

Paychecks

Atlanta drivers report fewer trips and thinner pay since Waymo arrived. Their union wants a 50¢ to $1 fee on each robotaxi ride, paid to drivers.

The catch: the gains are spread across lots of people. The costs land on drivers who know exactly who they are.

Who’s counting?

CASE

THE CLAIM: “FEWER CRASHES THAN HUMANS”



Waymo chose the roads and weather.
The human bar includes drunk and exhausted drivers.

The honest version: fewer crashes than the average human driver, on roads Waymo chose, by Waymo’s own count.

WHO COUNTS?

Mostly the company.

HOW OFTEN DO THEY NEED HELP?

Seven companies wouldn’t tell a senator.

DO PEOPLE TRUST IT?

Six in ten drivers would be afraid to ride (AAA, 2025).

Waymo failures: When it fails, it fails on camera

CASE

1. **School buses (Austin and Atlanta, fall 2025): no children were hurt.** It started in Atlanta. NHTSA launched its investigation in October after footage showed a Waymo maneuvering around a stopped school bus unloading children. Then Austin's school district logged the same thing again and again. The closest call: a Waymo drove past a stopped bus moments after a student crossed in front of it, while the student was still in the roadway.

2. **An illegal U-turn.** Police on a drunk-driving patrol watched a Waymo make an illegal U-turn right in front of them. They pulled it over, and there was nobody to ticket. California has since changed the law so police can cite the company.

3. **The child (Santa Monica, Jan 23, 2026: a child was hurt, with minor injuries.** During morning drop-off near Grant Elementary, the child ran across the street from behind a double-parked SUV towards the school and was struck by the Waymo. Waymo says the car slowed from about 17 mph to under 6 before contact. Afterward, the child "stood up immediately, walked to the sidewalk, and we called 911." Federal regulators opened an investigation.



Not a Waymo photo. Katy Warner, CC BY-SA 2.0.

The fix: more data and reinforcement learning? every fix is a new line on the score sheet: the car learned a lesson nobody wrote down until it failed on camera.

Waymo: reflect, then discuss

GROUP DISCUSSION

1. **Benefits vs. Job Loss.** Waymo reports far fewer serious crashes than human drivers. A trauma surgeon on NPR calls it a public health breakthrough. Drivers in Atlanta say their pay is already shrinking. So what should a city do?

2. **Responsible AI?** Slide 14 listed four things a company owes the street: explain its mistakes, know where it's safe, let others keep score, and answer when it's wrong. Which one has Waymo done best, and which worst? Use one of today's failures as your evidence.

3. **Long tail?** Waymo's safety numbers come mostly from ordinary miles, on roads and in weather it chose. Name one rare situation a robotaxi might meet in Bellingham. Do those safety numbers tell you anything about it? Would practice in a world model be enough proof for you?

4. **No driver?** Some women and disabled people say they feel safer with no stranger in the car. But one rider was stuck when men blocked her Waymo, and there was no driver to help her get away. Is an empty driver's seat safer or less safe for the person in the back? If it's the help button, who should answer it, and how fast?

5. **Keep, fix, or drop?** Waymo wants to operate in Bellingham. Yes, no, or yes with conditions?

1. Groups, 7 min.
Everyone does #5, plus the one I assign.

Pick a leader.

2. Whole class, 5 min.
One leader per question.

ATTENDANCE BEFORE 4PM PLEASE!

Upload your notes to Canvas right after class.

Wednesday: Lab 7, The Bellingham Pitch. You work for a robotaxi company and pitch Bellingham's city council. Then we debate: should Seattle let robotaxis take over its taxi and rideshare work?

Friday 11/6, 11:59 p.m.: Lab 7 writeup and Chapter 7 quiz.